Essential Model Documentation

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Motivation

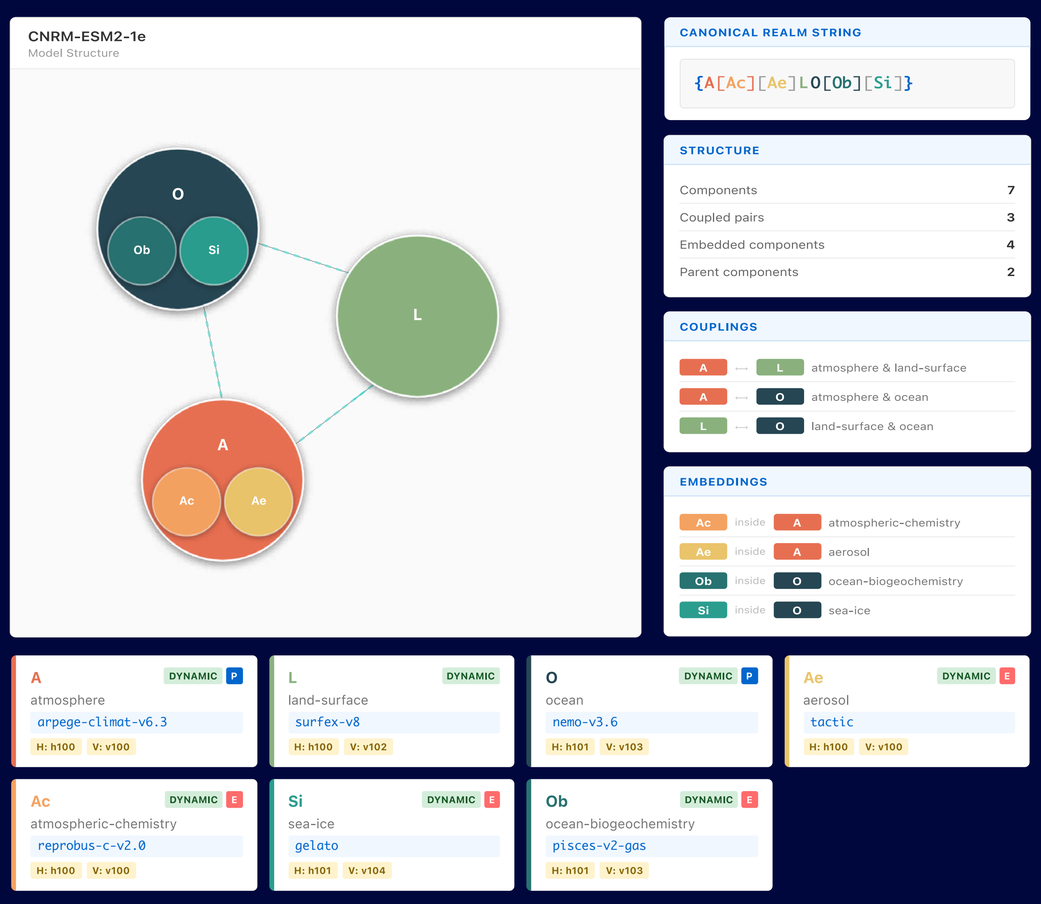
The Essential Model Documentation (EMD) provides a clear and standardised description of an Earth System Model, including its components, simulated processes, and configuration. This enables users to assess whether a model is appropriate for a given research question and to compare models on a consistent basis.

EMD is a mandatory requirement for the CMIP7 project (Dunne et al., 2025) and ensures transparent model documentation while minimising the reporting burdens for modelling groups.

What does the EMD allow us to do?

Knowing how a climate model is built — which parts talk to each other and how they are arranged, allows researchers to understand why two models are likely to produce different output.

By collecting this information we are able to compare different model configurations and use them like a fingerprint to identify similar models.



What is in it?

STAGE 1

g-cells
Horizontal Grid Cells

Individual tiles forming a model's horizontal domain — their shape, resolution, geographic coverage, and origin. These cells define where all variables are computed. Understanding resolution is critical: finer grids resolve local features and weather systems, while coarser grids run faster but miss regional details.

↗ region CV ↗ grid type CV

STAGE 2

h-grids
Horizontal Computational Grid

How g-cells are arranged into a computational grid, including their staggering and how variable types are distributed. This arrangement creates a collection of reusable sub-grids that can be referenced by multiple models. The staggering type (Arakawa-A, C, etc.) affects numerical accuracy.

→ g-cells ↗ variable type

v-grids
Vertical Grid

Layer structure in the vertical dimension — number of levels, coordinate system (pressure, terrain-following, hybrid), and total depth/height extent. Coarse vertical resolution may miss boundary layer processes or stratification important for your research question.

↗ coordinate CV

STAGE 3

model_component
Model Component

A single scientific sub-model — atmosphere, ocean, land surface, or sea ice. Records what it simulates, its version, and the grids it runs on. Knowing which physical processes are included (or excluded) is essential to understand model strengths and limitations for your application.

→ h-grid → v-grid → component family ↗ DOI / references

STAGE 4

model
Climate Model

A complete Earth system or climate model. Records which components are active (dynamically simulated), prescribed, or omitted; how they exchange information (coupling strategy); and configuration details. This tells you what feedbacks are included and which processes may be artificially fixed, affecting model behavior and uncertainty.

→ model_component → ESM family ↗ DOI / references

SUPPORTING

model_family
Component Family

Groups versions of the same component code base and tracks shared scientific heritage across configurations. Helps you trace evolution of a component's code, identify what changed between versions, and understand relationships between models.

↗ institution

model_family
ESM Family

Groups versions of the same coupled model system. Allows you to trace how a model evolved across CMIP generations, compare variants at different resolutions, and identify shared scientific heritage to understand systematic differences in results.

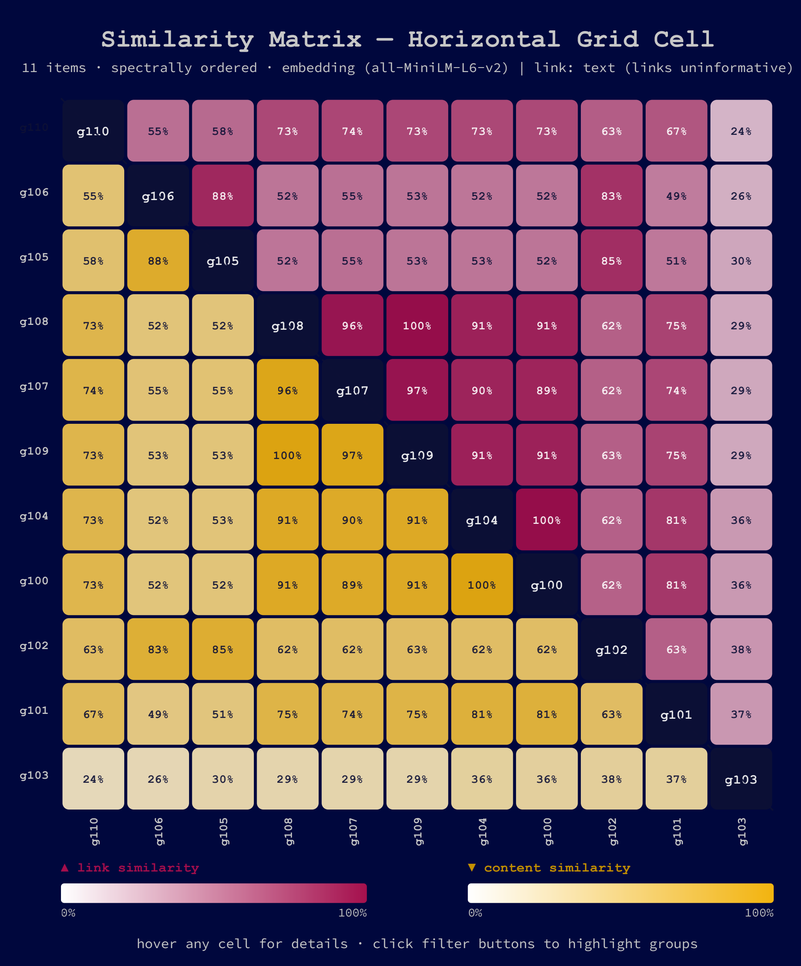
↗ institution

Comparing Similarity

Link-based similarity looks at the external URI references in the EMD content and compares them using Jaccard similarity. In simple terms, this shows which items point to the same institutions, coordinate systems, or regions, helping us identify structural connections between them.

We then analyse the remaining fields and use text embeddings generated with all-MiniLM-L6-v2 (Reimers & Gurevych, 2019) to explore the differences between files. This tells us how semantically similar the content is - even if the structure or wording is different.

The results of these are shown as a similarity graph (an adjacency matrix). We apply spectral sorting based on the eigenvector of the second-smallest eigenvalue of the graph Laplacian (Fiedler, 1973) to rearrange the graph and make patterns easier to see. This is used to identify clusters of similar items with darker groups situated on the diagonal.



Automation and EMD Review

Storing information in JSON-LD (Linked Data) format supports a cleaner software architecture, with less duplication, fewer errors, and reduced ambiguity at the source. It also allows us to reuse definitions across projects and automate much of the workflow — from collecting modelling centre information through GitHub forms, to running automated workers that validate and test submissions (ESGVOC + Pydantic).

New data files are generated automatically and submitted to GitHub together with a validation report for reviewers. This level of automation significantly reduces the effort required to assess submissions, as much of the technical validation is completed in advance.

However, human review remains essential. We rely on people to sense-check the content and assess the suitability of free-text fields.
If you are likely to engage with the EMD, we encourage you to join our community reviewer pool — please get in touch: email address shown at the bottom left.

How Can I See It?

The EMD can be accessed from either the GitHub pages link, or the URL below. This is also in the QR code.

https://emd.mipcvs.dev/docs

emd@wcrp-cmip.org